

Fatima Jinnah Women University, Rawalpindi

CC LAB_1

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Submitted By	Dua Malik
Reg. No	FA24B1-SE-034
Subject	Cloud Computing
Git Hub username	dua-malik612
Session	2024-28
Section	B
Submitted To	Sir Waqas Saleem

Getting Started: Repository Folder Structure

Explanation: Created the required Labs/Lab01/screenshots/ directory inside the CC repository to organize and store all evidence files for this lab.

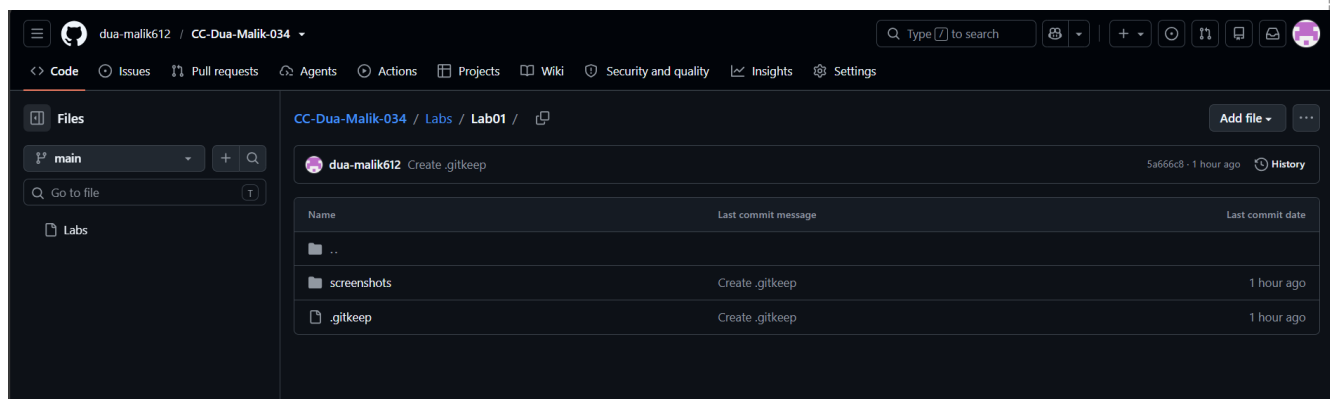


Figure 0.1: Created Repository Folder Structure

Task 1: Create and Configure Your GitHub Account

Explanation: Created a personal GitHub profile using a professional username (dua-malik612) based on my actual name, for lab identification and project tracking.

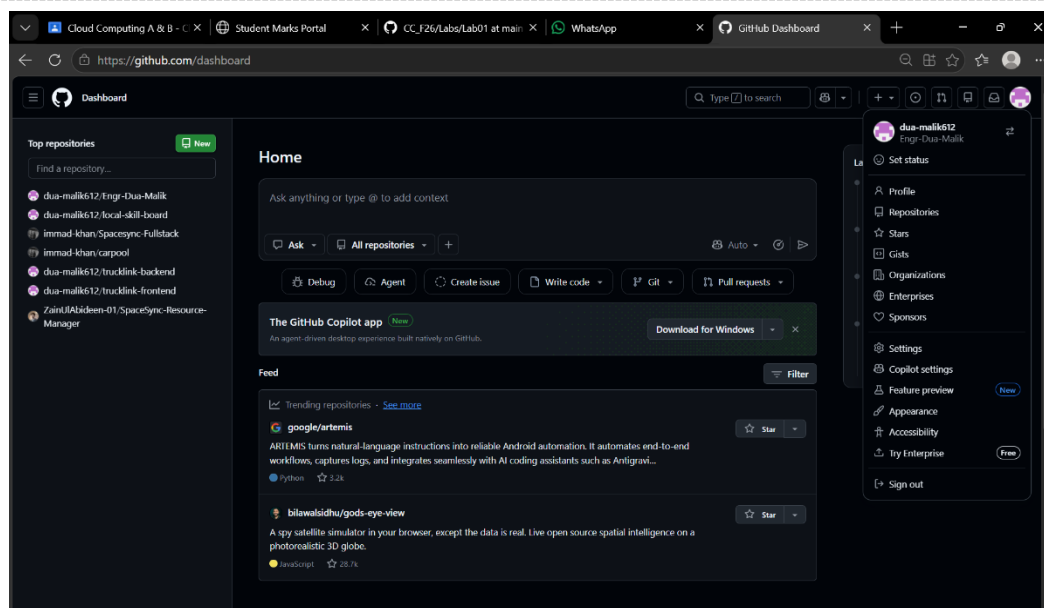


Figure 1.1: GitHub Account Creation & Username Setup

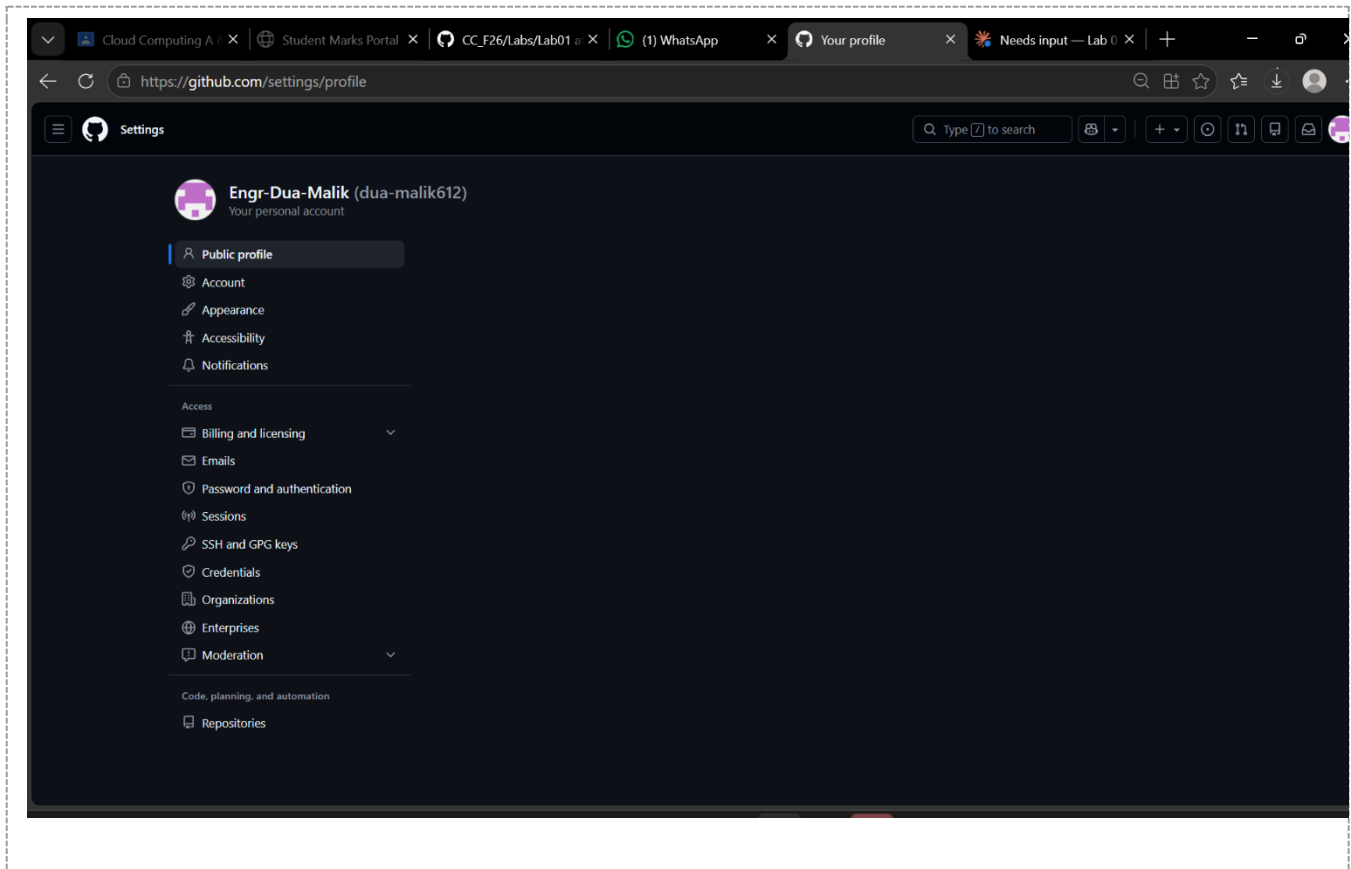


Figure 1.2: *Configured Public Profile Settings*

Task 2: Register in the Student Marks Portal

Explanation: Authenticated the GitHub account with the Student Marks Portal via GitHub sign-in and submitted the enrollment request for the Cloud Computing course.



Figure 2.1: *Student Marks Portal Login Page*

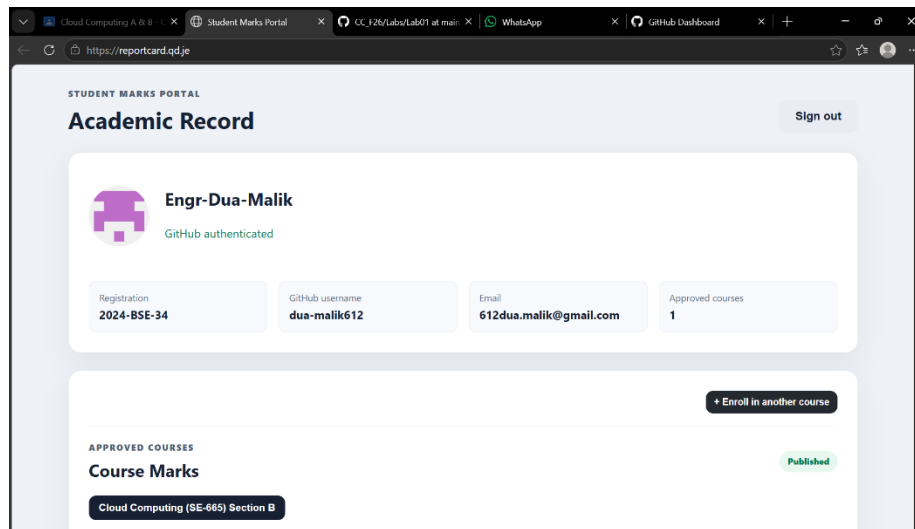


Figure 2.2: *Authenticated GitHub Profile Details inside Portal*

+ Enroll in another course			
APPROVED COURSES			
Course Marks			
Published			
Cloud Computing (SE-665) Section B			
Class Participation			
ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Class Participation	0	10	0.0%
Total	0	10	0.0%

Figure 2.3: *Course Enrollment Status Submitted / Pending Approval*

Task 3: Environment Prerequisites Verification

Explanation: Verified all system requirements, including the installation of VMware Workstation Pro, availability of adequate host disk space, and downloading of the Ubuntu Server ISO.

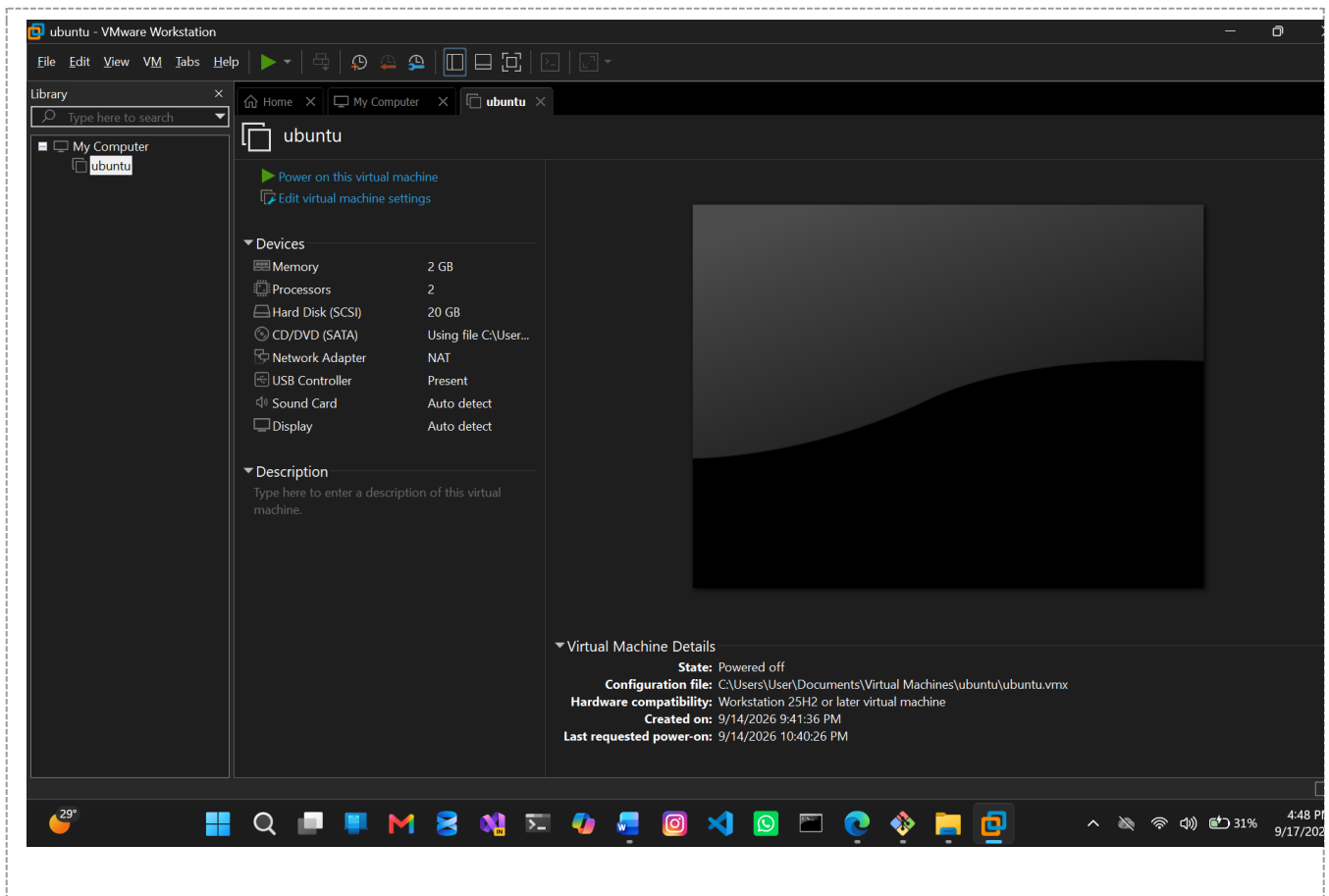


Figure 3.1: *VMware Workstation Pro Home Screen*

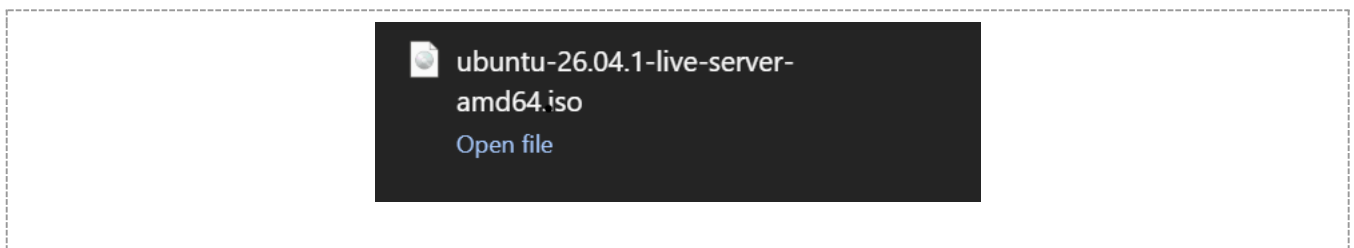


Figure 3.2: *Downloaded Ubuntu Server ISO File Confirmation*



Figure 3.3: *Available System Storage Space on Host Drive*

Task 4: Virtual Machine Creation in VMware

Explanation: Initialized a new virtual machine using the Typical (recommended) configuration wizard and mapped the downloaded ISO file to boot into the Ubuntu Server setup.



Figure 4.1: *New Virtual Machine Creation Wizard Launch*



Figure 4.2: *Typical Configuration Selection*

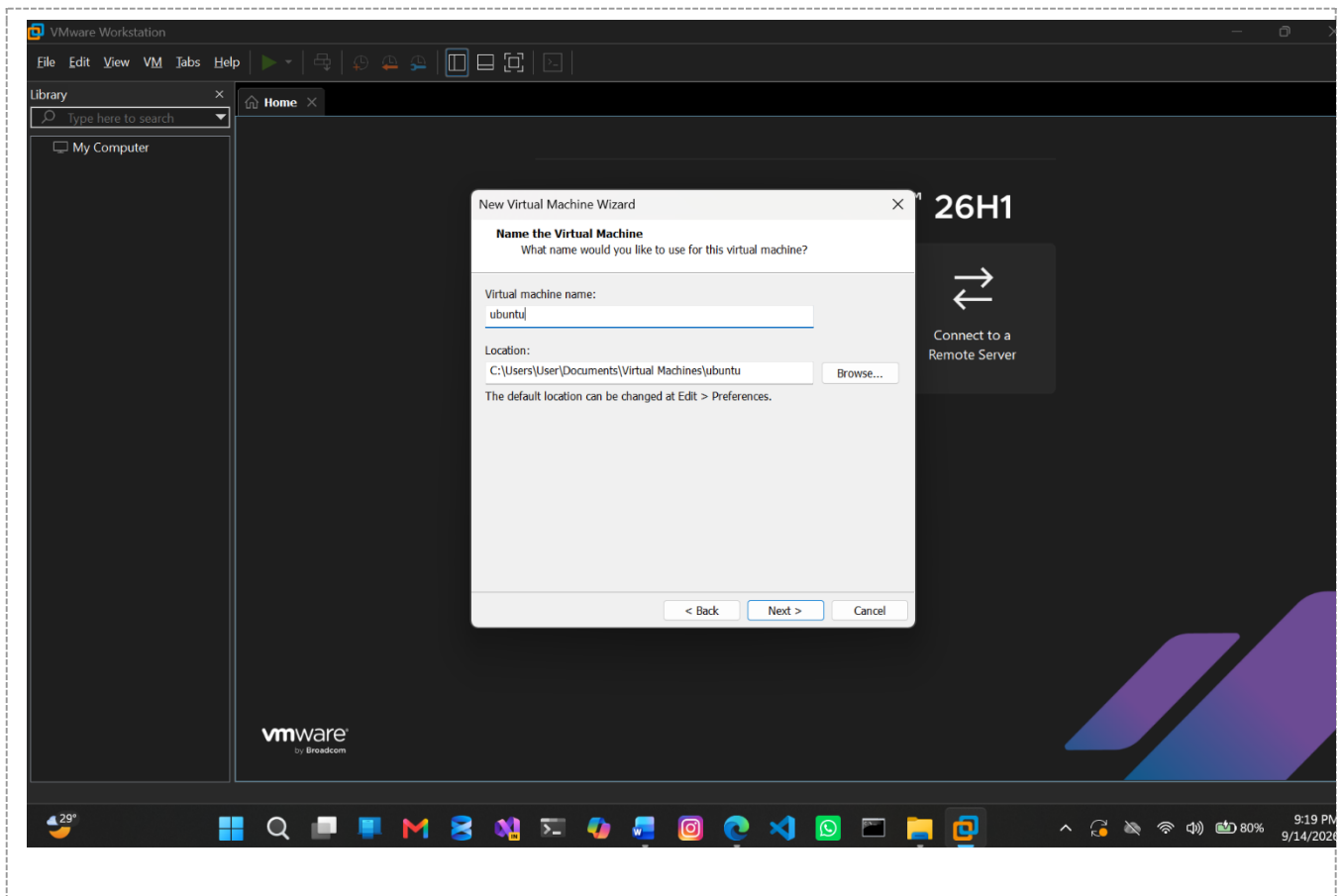


Figure 4.3: *Selected Ubuntu ISO Path*

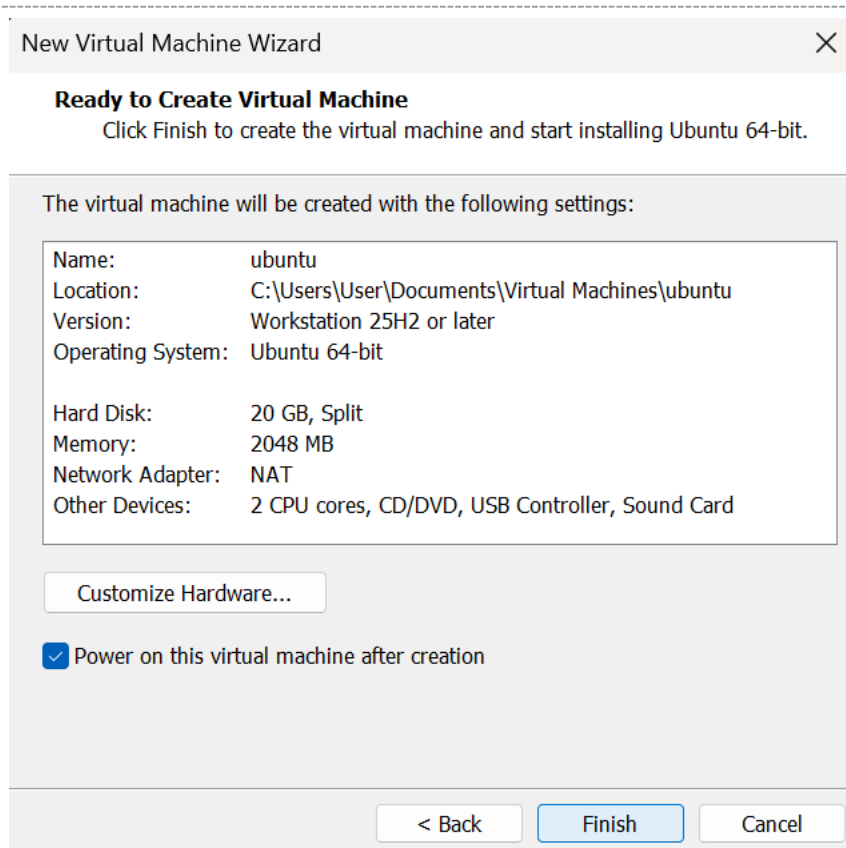


Figure 4.4: *Final Virtual Machine Hardware Summary*



Figure 4.5: *GNU GRUB Boot Menu for Ubuntu Server*

Task 5: Language Selection

Explanation: Selected English as the system installation language for the Ubuntu Server installer.

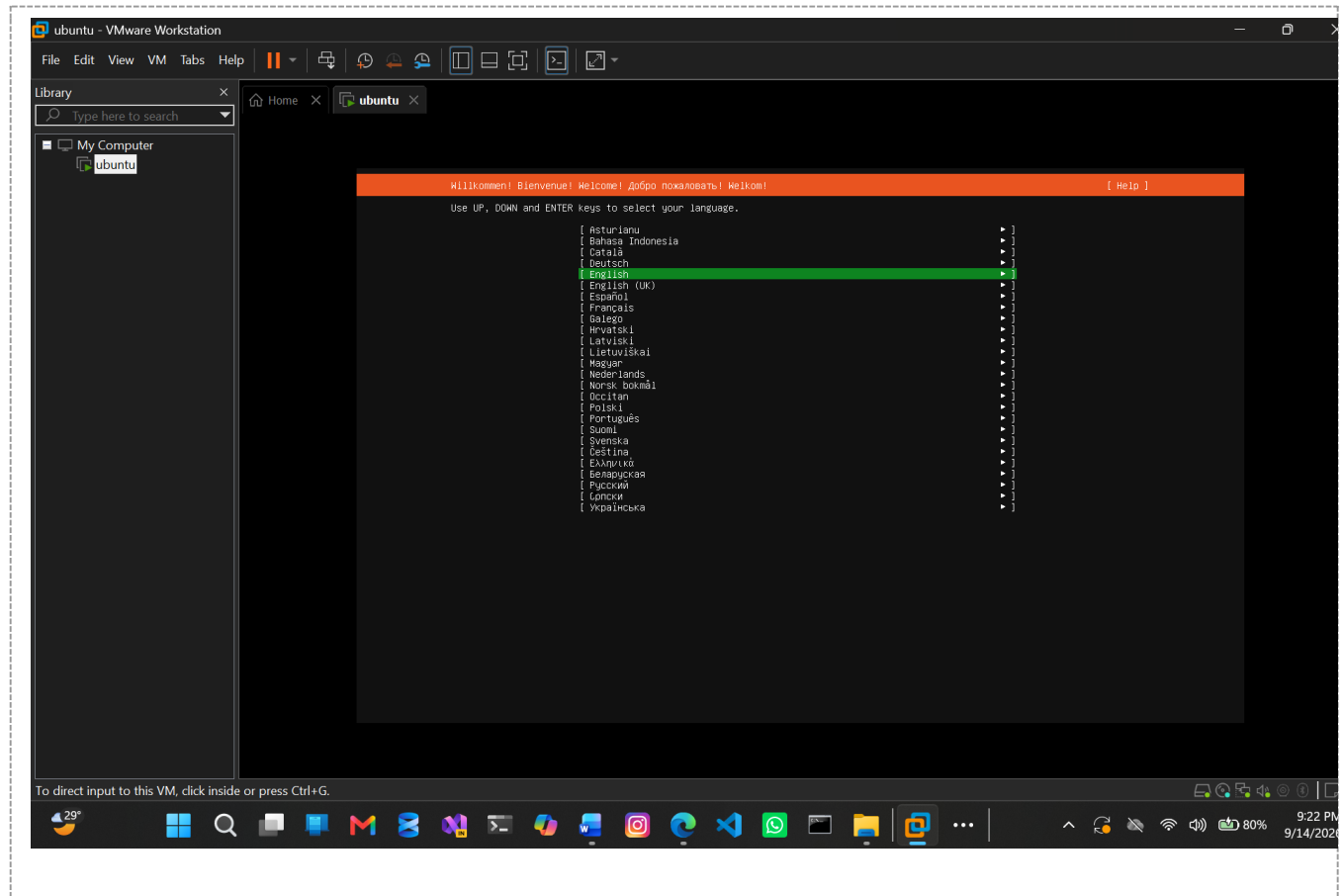


Figure 5.1: *Ubuntu Server Installer Language Selection Screen*

Task 6: Keyboard Configuration

Explanation: Configured the default English (US) keyboard layout and variant to match the host system's keyboard hardware.

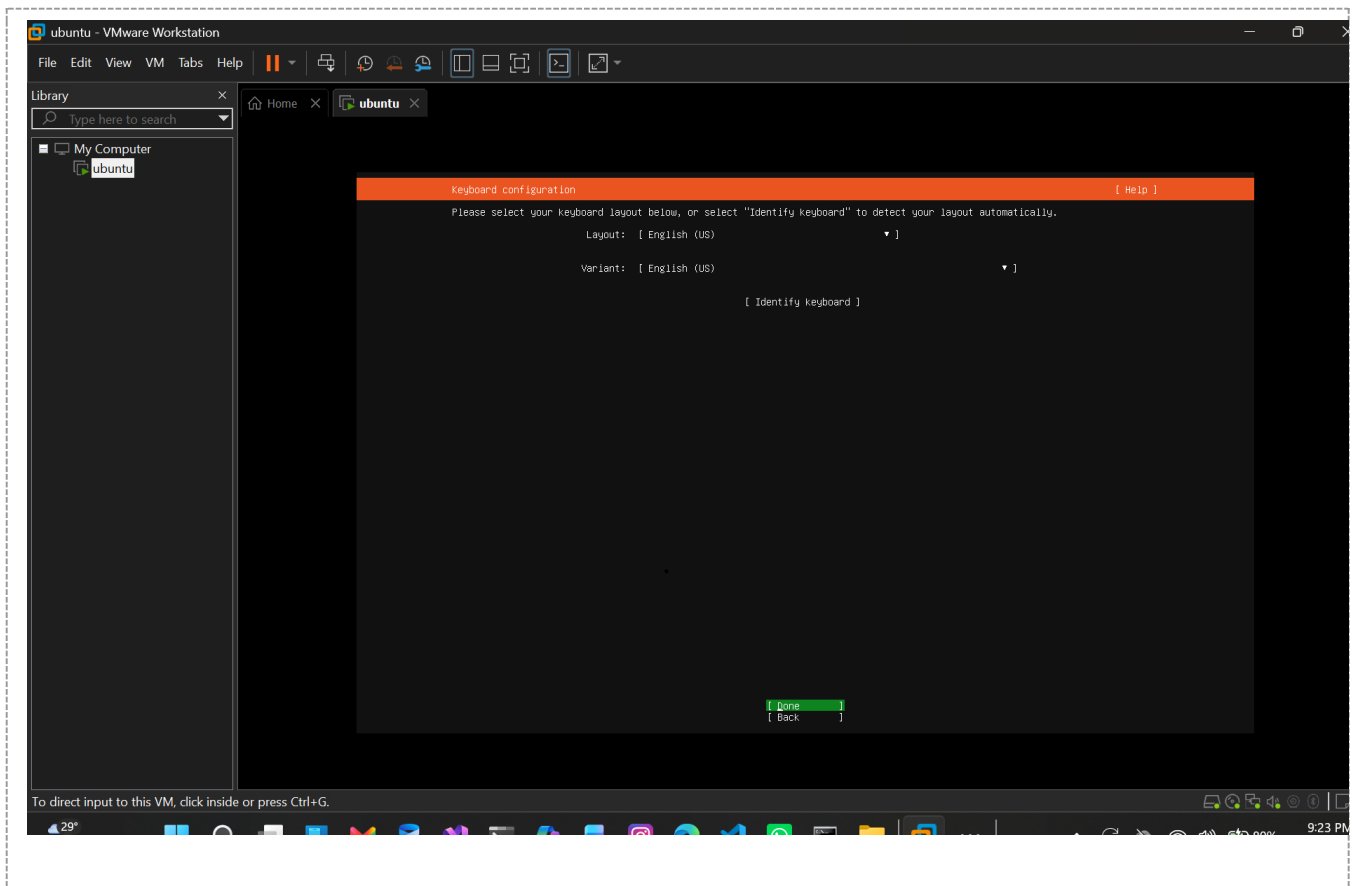


Figure 6.1: *Selected English (US) Keyboard Layout*

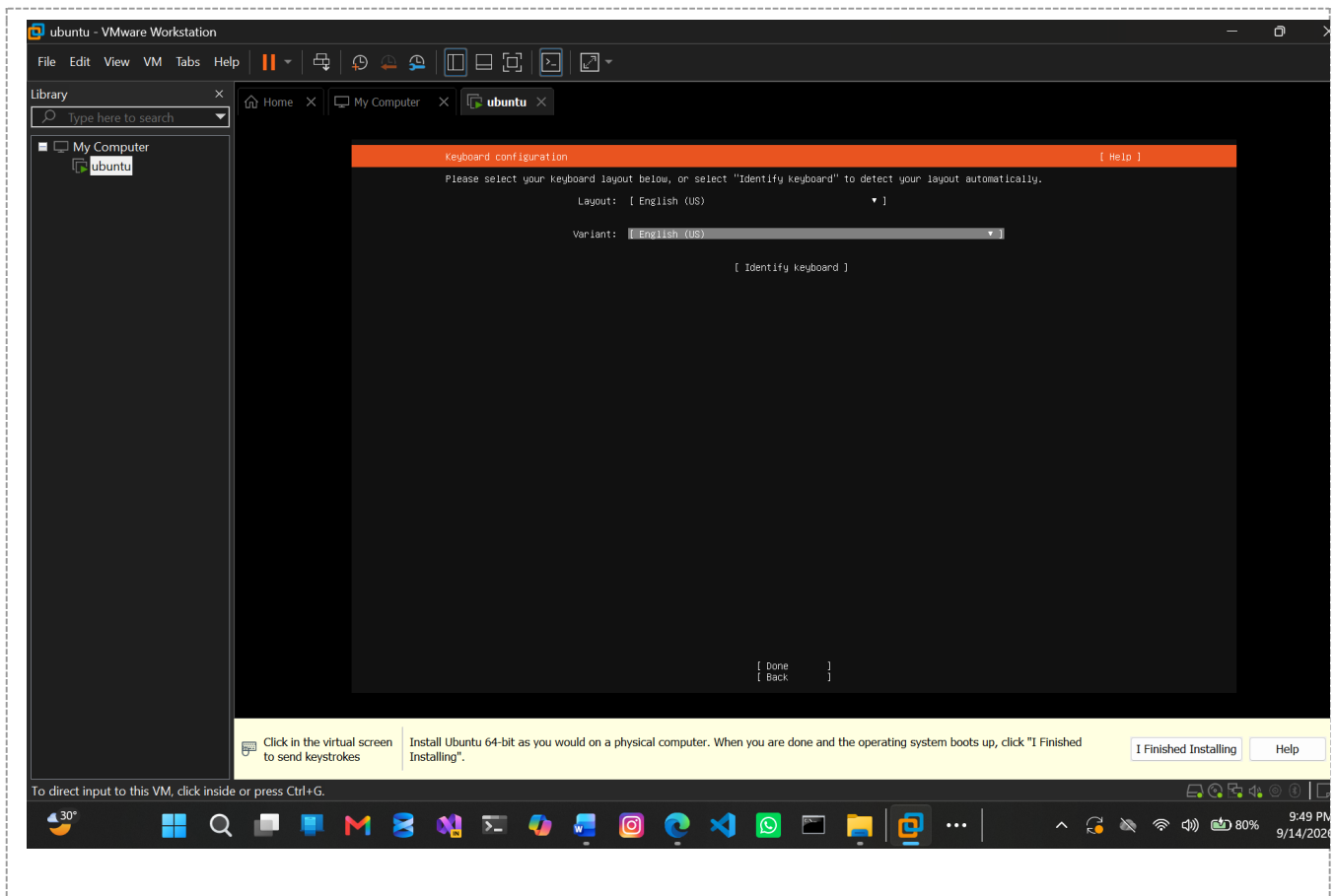


Figure 6.2: *Selected Keyboard Variant*

Task 7: Installation Type Selection

Explanation: Chose the normal (base) Ubuntu Server installation option to provide the standard set of server utilities.

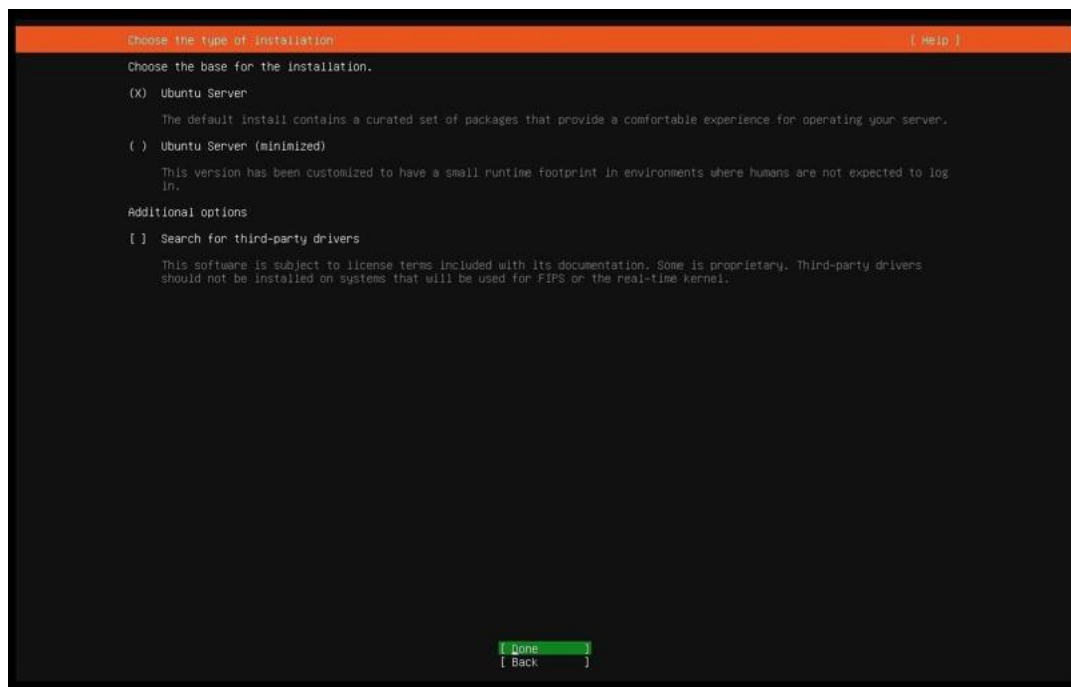


Figure 7.1: *Base Ubuntu Server Installation Option Selected*

Task 8: Network Configuration

Explanation: Verified detection of the network interface (ens33) and confirmed that a dynamic IP address was assigned automatically via DHCP.

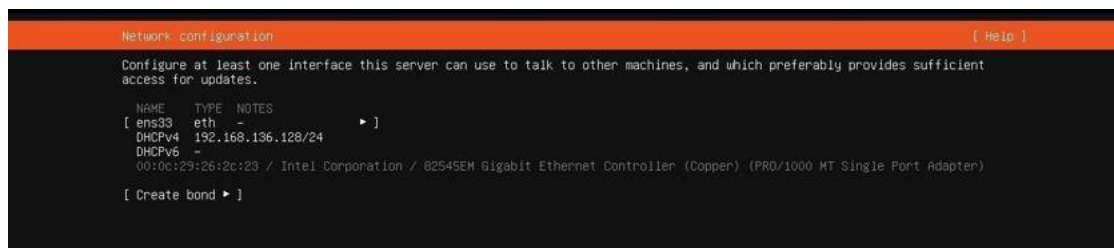


Figure 8.1: *Detected Ethernet Interface*

Task 9: Guided Storage Layout Setup

Explanation: Configured the guided storage layout to use the entire virtual disk, set up as an LVM (Logical Volume Manager) group.



Figure 9.1: Guided Storage Configuration Screen

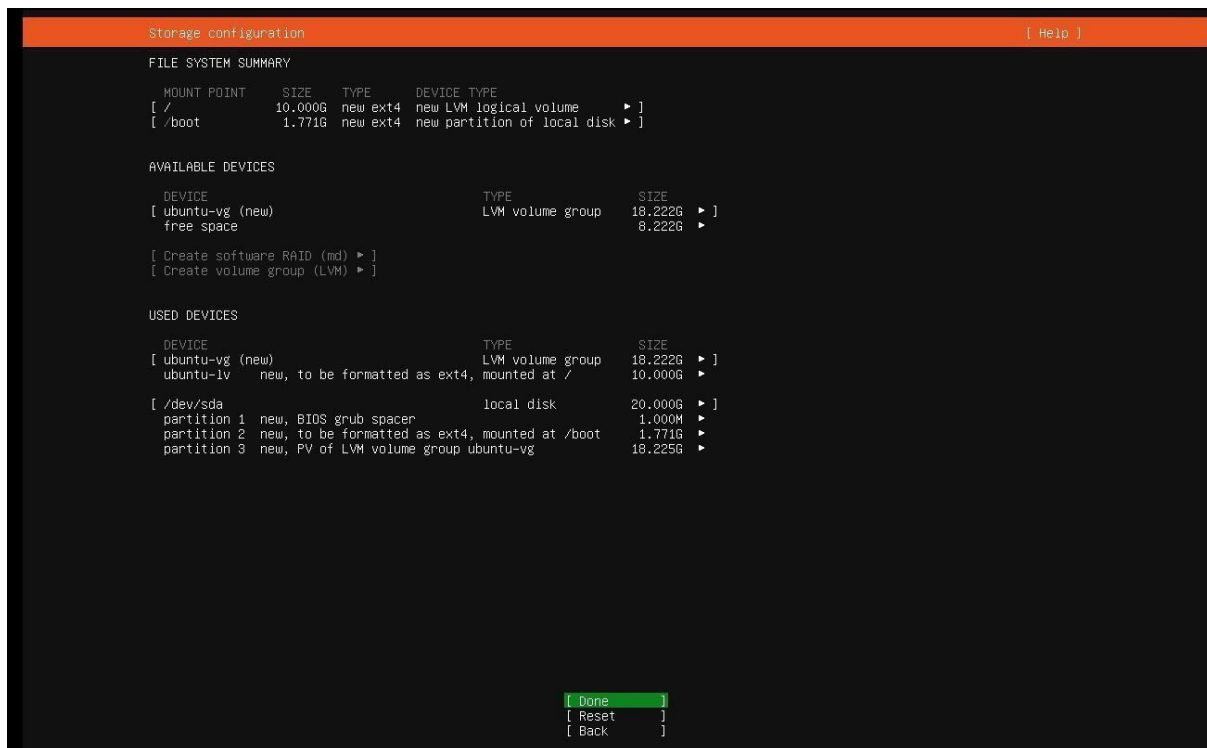


Figure 9.2: Resulting File System Summary and Volume Layout

Task 10: Select the Installation Disk

Explanation: Mapped the installation storage to the allocated VMware virtual hard disk (/dev/sda) and confirmed its size.

Guided storage configuration

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

[/dev/sda local disk 20.000G ▾]

[X] Set up this disk as an LVM group

[] Encrypt the LVM group with LUKS

Passphrase:

Figure 10.1: Target Local Disk Selection

Storage configuration

FILE SYSTEM SUMMARY

MOUNT POINT	SIZE	TYPE	DEVICE	TYPE
[/	10.000G	new ext4	new LVM logical volume	►]
[/boot	1.771G	new ext4	new partition of local disk	►]

AVAILABLE DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	18.222G ►]
free space		8.222G ►
[Create software RAID (md) ►]		
[Create volume group (LVM) ►]		

USED DEVICES

DEVICE	TYPE	SIZE
[ubuntu-vg (new)	LVM volume group	18.222G ►]
ubuntu-lv	new, to be formatted as ext4, mounted at /	10.000G ►
[/dev/sda	local disk	20.000G ►]
partition 1	new, BIOS grub spacer	1.000M ►
partition 2	new, to be formatted as ext4, mounted at /boot	1.771G ►
partition 3	new, PV of LVM volume group ubuntu-vg	18.225G ►

Figure 10.2: Disk Allocation and File System Mount Summary

Task 11: Review the Partitions

Explanation: Reviewed the proposed partition layout, including the /boot partition and LVM logical volume mapping, before proceeding to write changes to disk.

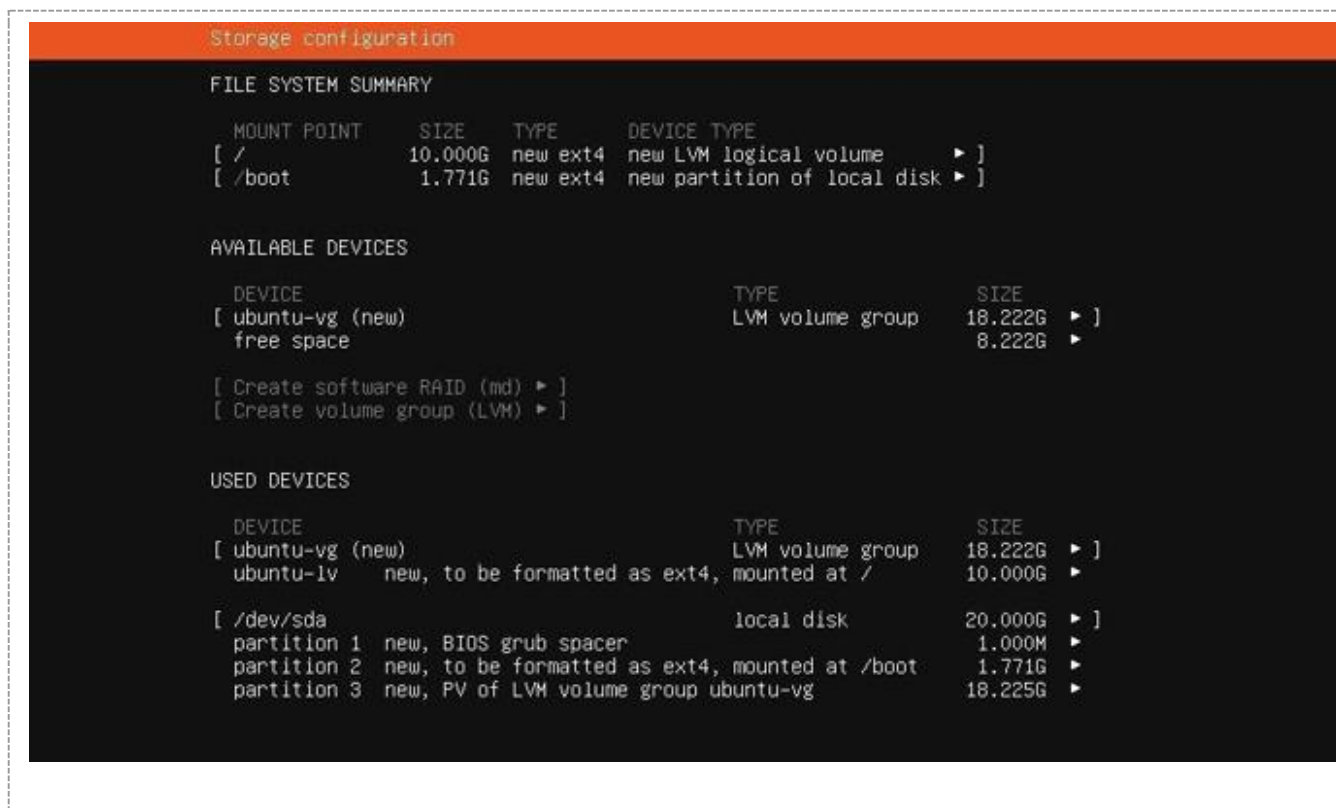


Figure 11.1: Proposed Storage Partition Table Summary

Task 12: Confirm the Storage Changes

Explanation: Read the destructive-action warning carefully and confirmed the change, allowing the installer to format the VMware virtual disk partitions.

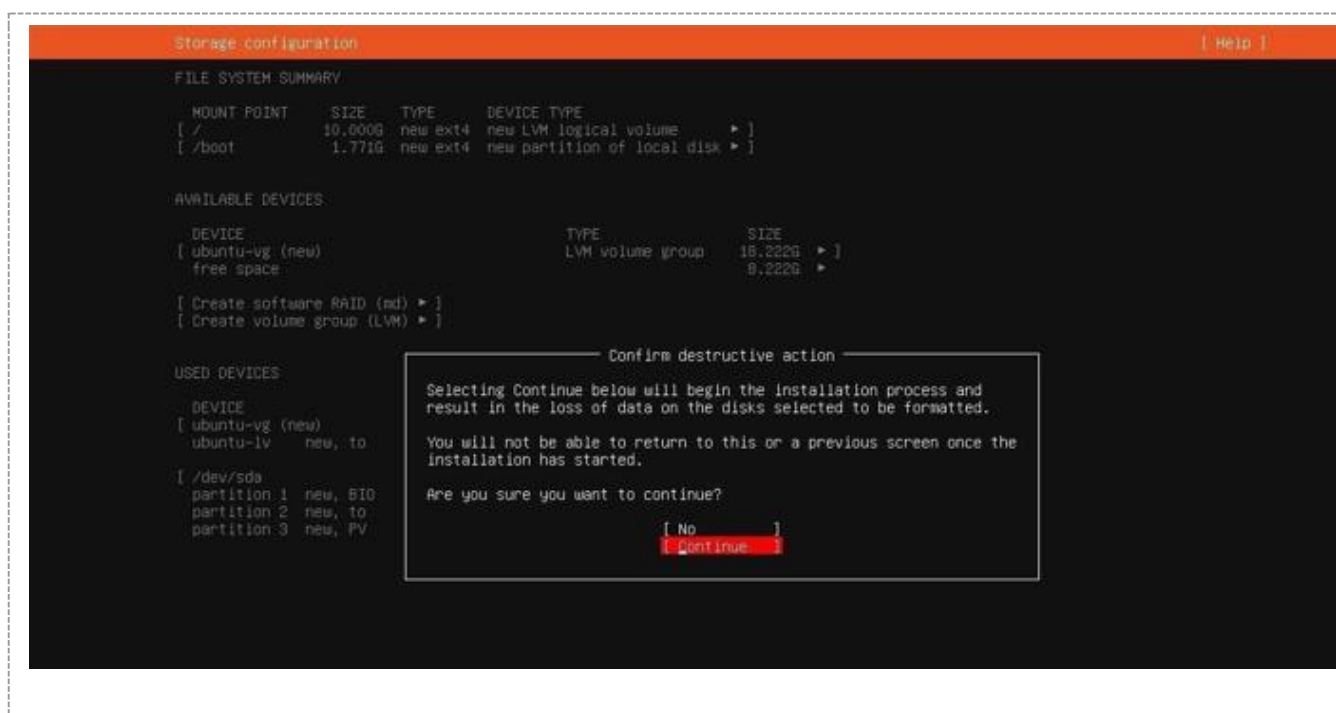


Figure 12.1: Destructive Action Warning Prompt

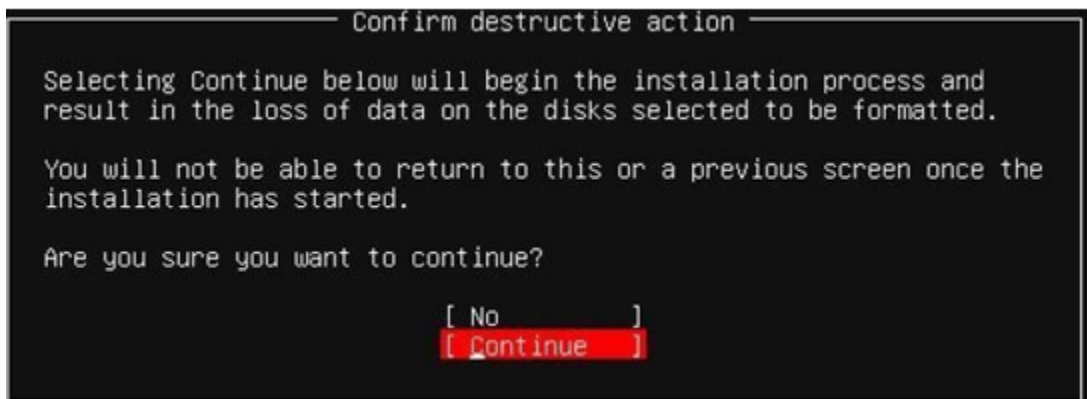


Figure 12.2: *Confirmation Choice Selected (Continue)*

Task 13: Configure the Ubuntu Profile

Explanation: Configured the server profile with my complete actual name, set the server hostname to 'ubuntu', and set my exact GitHub username (dua-malik612) as the Ubuntu username.

A terminal window with a black background and white text. It shows the prompt "Your name:" followed by the text "Dua Malik" entered in the input field.

Figure 13.1: *Entered System User Name*

A terminal window with a black background and white text. It shows the prompt "Your servers name:" followed by the text "ubuntu" entered in the input field. Below the input field, it says "The name it uses when it talks to other computers."

Figure 13.2: *Configured Server Hostname*

A terminal window with a black background and white text. It shows the prompt "Pick a username:" followed by the text "dua-malik612" entered in the input field.

Figure 13.3: *Assigned System Username*

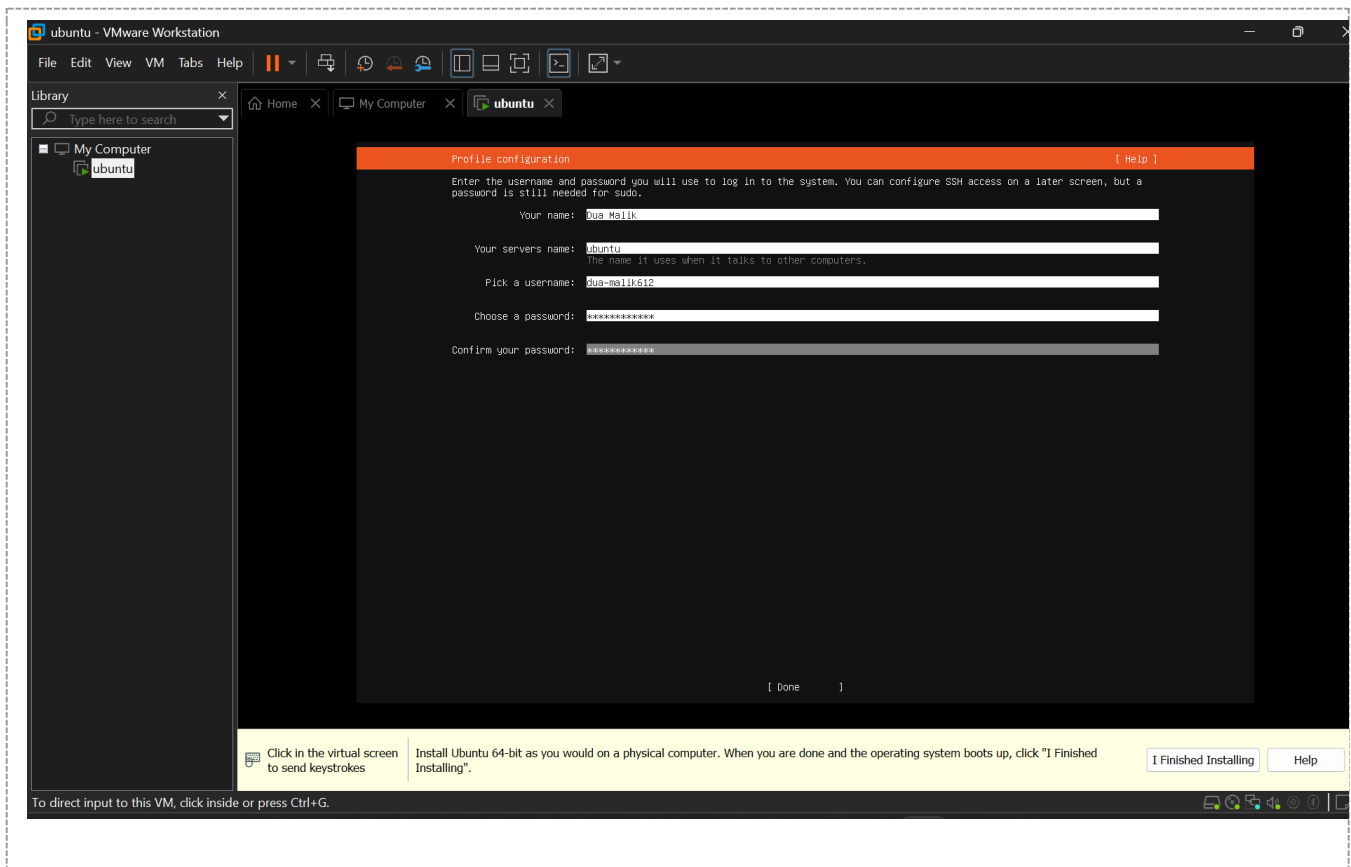


Figure 13.4: *Completed Profile Configuration Screen*

Task 14: Complete the Software Installation

Explanation: Monitored the real-time system file extraction, core package installation, and OpenSSH server deployment during the installation process.

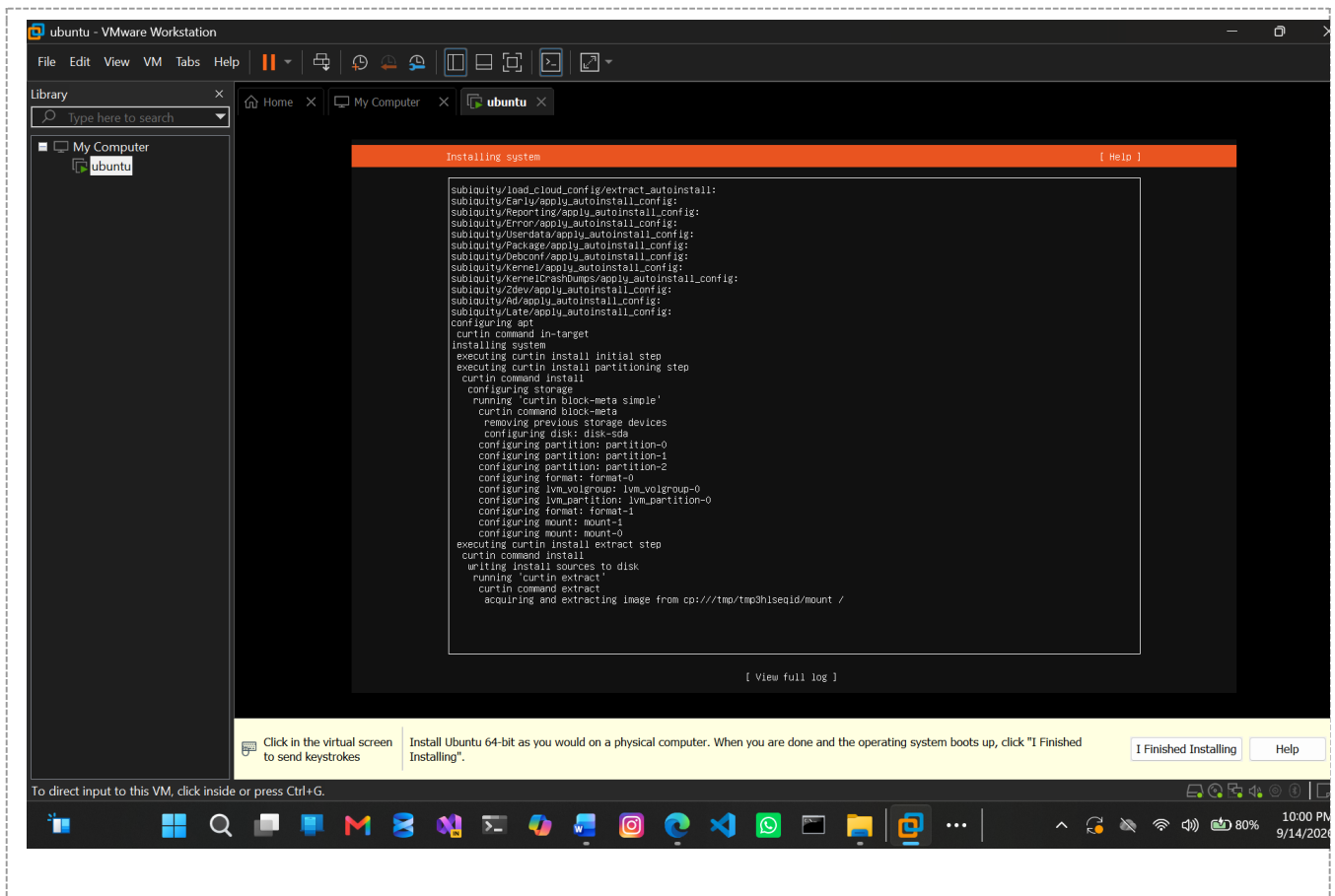


Figure 14.1: System Files Extraction and Installation Progress Log

Task 15: Reboot and Verify the Ubuntu Identity

Explanation: Completed the system reboot, reached the login prompt, and successfully signed in to verify that the terminal prompt follows the required format <github-username>@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address

Explanation: Executed the `ip addr` command to retrieve the active network configuration and noted the server's assigned IPv4 address.

```
dua-malik612@ubuntu:~/mylabs$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:80:63:db brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamename enx000c298063db
    inet 192.168.157.129/24 metric 100 brd 192.168.157.255 scope global dynamic ens33
        valid_lft 1153sec preferred_lft 1153sec
    inet6 fe80::20c:29ff:fe80:63db/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Figure 16.1: Output of `ip addr` Command

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:80:63:db brd ff:ff:ff:ff:ff:ff
    altnamename enp2s1
    altnamename enx000c298063db
    inet 192.168.157.129/24 metric 100 brd 192.168.157.255 scope global dynamic ens33
        valid_lft 1153sec preferred_lft 1153sec
    inet6 fe80::20c:29ff:fe80:63db/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Figure 16.2: Highlighted IPv4 Address on `ens33` Interface

Task 17: Connect to Ubuntu Server from Windows

Explanation: Established a remote SSH connection to the Ubuntu server from Windows Command Prompt, accepted the host authenticity key fingerprint, and verified remote terminal prompt access.

```
dua-malik612@192.168.157.129's password:
```

Figure 17.1: Typed SSH Connection Command

```
The authenticity of host '192.168.157.129 (192.168.157.129)' can't be established.
ED25519 key fingerprint is SHA256:bsUKWWh71bTS7DgzVxygPYDUG2TrIa1CD+EYERFbq2w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.157.129' (ED25519) to the list of known hosts.
```

Figure 17.2: Host Fingerprint Key Acceptance

```
dua-malik612@192.168.157.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)
```

Figure 17.3: Remote User Password Authentication Input

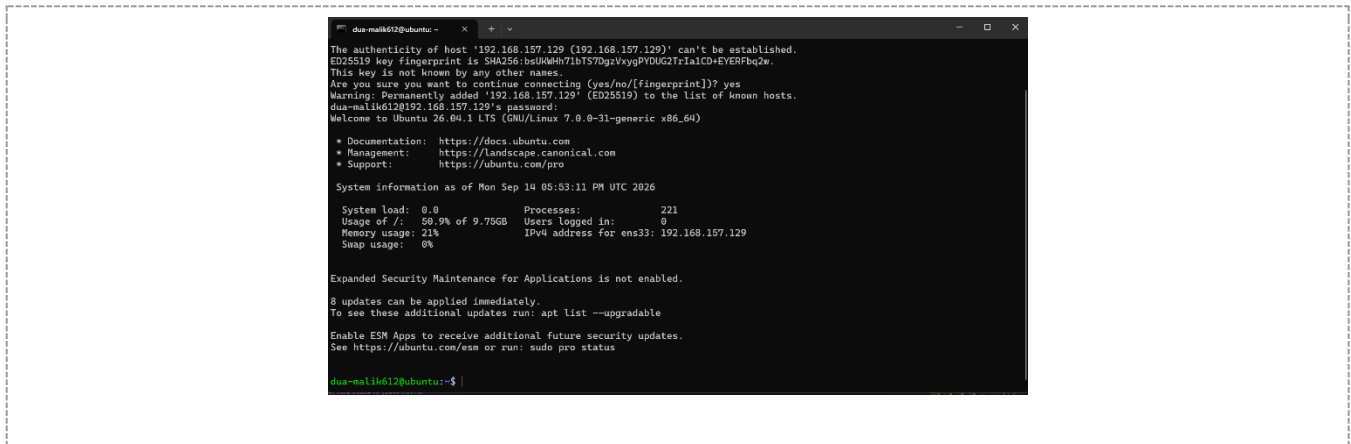


Figure 17.4: Established SSH Remote Session Showing Active Remote Prompt

- ✓ README.md remains inside Labs/Lab01/.
- ✓ The instructor-provided images/install-ubuntu-server/ directory remains unchanged.
- ✓ Lab1_Solution.pdf was created by me and placed inside Labs/Lab01/.
- ✓ Every original evidence screenshot is inside Labs/Lab01/screenshots/.
- ✓ My GitHub profile contains my complete actual name.
- ✓ My GitHub username is professional and suitable as an Ubuntu username.
- ✓ I used the same GitHub account for the Student Marks Portal.
- ✓ I entered my correct registration number, course code, and section.
- ✓ Ubuntu Server was installed inside VMware Workstation Pro.
- ✓ My Ubuntu username exactly matches my GitHub username.
- ✓ My Ubuntu server name is ubuntu.
- ✓ My local terminal prompt shows <github-username>@ubuntu.
- ✓ I identified the correct Ubuntu Server IPv4 address.
- ✓ I successfully connected to Ubuntu Server from Windows using SSH.
- ✓ My remote SSH prompt shows <github-username>@ubuntu.
- ✓ Every screenshot uses the exact required filename.

-----THE END-----